

Yueran (Hannah) Sun

Master's Student in Data Science | University of Washington

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EDUCATION

2025-Present	University of Washington — M.S. in Data Science ‣ Expected graduation: Mar 2027 ‣ GPA: 4.0/4.0	Seattle, WA
2021-2025	University of Michigan — B.S. in Data Science ‣ Minor: Mathematics ‣ GPA: 3.71/4.0 (University Honors)	Ann Arbor, MI

RESEARCH INTERESTS

Human-Computer Interaction, Human-Centered AI, AI and Society, Accessibility, Information Seeking, AI Safety, Multi-Agent Systems, Mechanistic Interpretability, Computational Humanities, Multimodal Information Retrieval

RESEARCH EXPERIENCE

Present Sep 2026	Research Fellow , Supervised Program for Alignment Research <i>Mentor: Emilio Barkett</i> ‣ Analyzing LLM judgment revision during collaborative problem solving to identify naturally emerging social influence and conformity-like behavior. ‣ Developing counterfactual and mechanistic interpretability analyses to identify causal drivers and internal mechanisms of judgment revision, including social commitment and psychological reactance.	Berkeley, CA (Remote)
Present Sep 2025	Research Collaborator , UW Language Accessibility Research Lab <i>PI: Prof. Lucy Lu Wang</i> ‣ Annotating chronic illness discussions on Reddit to identify accessibility barriers and solutions related to symptom management, with planned follow-on data analysis. ‣ Evaluated AI-assisted scholarly reading interfaces in a classroom study comparing baseline reading, on-demand augmentation, and elaborative interrogation, assessing immediate and delayed comprehension, cognitive load, and reader preferences. ‣ Examined gaps between automated readability metrics, AI-generated adaptations, and the plain-language translation priorities of disability advocacy organizations.	Seattle, WA
Present Sep 2025	Research Collaborator , UW Lab for Computing Cultural Heritage <i>PI: Prof. Benjamin Charles Germain Lee</i> ‣ Engaging in an independent study on multimodal linking of archival photographs and oral-history transcripts to support interactive exploration of derived entities and relationships. ‣ Developed a CLIP- and DBSCAN-based computer vision pipeline for clustering, retrieving, and semantically labeling reprinted photographs across 1.5M historical newspaper images.	Seattle, WA
Jul 2026 Jun 2026	Graduate Research Assistant , Humanities Data Science Summer Institute American Institute of Physics Team, <i>PI: Prof. Benjamin Charles Germain Lee</i> ‣ Mentored four undergraduate researchers to improve discovery across American Institute of Physics archives, transforming 1,700+ oral histories into knowledge graphs and linking unnamed collaborators to external knowledge bases.	Seattle, WA
May 2024 Nov 2023	Undergraduate Research Assistant , MobiDrop Co., Ltd. <i>Mentor: Dr. Guantao Zheng</i> ‣ Pretrained and evaluated a Transformer-based model for single-cell gene expression on 300,000 human blood cells, automating GPU training and checkpointing while achieving a 6% reduction in validation error on masked gene expression prediction.	Shanghai, China (Remote)

Aug 2023 Jul 2023	Undergraduate Research Assistant , Macao Polytechnic University <i>Mentor: Dr. Patrick Pang</i> Adapted BERTopic for topic modeling of Mandarin Chinese patient–physician discussions from an online healthcare consultation platform.	Macao SAR
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INDUSTRY EXPERIENCE

Aug 2025 Jun 2025	Software Engineer Intern , Ternary <i>Supervisor: Kyle Rattet</i> Enhanced the existing cloud cost prediction system with time-series forecasting to improve cost projections and deliver interactive forecasts through client dashboards.	Seattle, WA (Remote)
Dec 2024 Jan 2024	Student Developer , University of Michigan Multidisciplinary Design Program ProQuest Team, <i>Mentors: Prof. Sindhu Kutty, John Dillon, Dan Hepp</i> Developed an MLOps pipeline using large language models, Gaussian mixture models, and computer vision to segment historical newspapers into article-level structure, reducing segmentation costs by 50% compared to manual processes.	Ann Arbor, MI
Aug 2024 May 2024	NLP Algorithm Intern , Pachira Information Technology Co., Ltd. <i>Supervisor: Dr. Chenglong Ma</i> Implemented warning-light query handling, multi-intent recognition, and refusal logic for a RAG-based in-car voice assistant for Toyota vehicles.	Zhuhai, China

PUBLICATIONS

C = Conference S = In Submission

- [C.1] **Beyond Readability Metrics: Plain Language Priorities in Disability Advocacy Organizations**
Kumar, A., Glazko, K., **Sun, Y.**, Harniss, M., Wang, L. L., & Mankoff, J.
ACM FAccT 2026
- [S.1] **Easy Reading Is Not Deep Reading: A Classroom Study of AI-Assisted Elaborative Interrogation on Reader Comprehension**
Kumar, A., **Sun, Y.***, Memon, S. A.*, & Wang, L. L.
*Under review, ACM CHI 2027 — *Equal contribution*
- [S.2] **Viral Images: Identifying Reprintings within 1.5 Million Photographs in Chronicling America**
Buccalon, B., **Sun, Y.**, & Lee, B. C. G.
Under review, Computational Humanities Research 2027

TEACHING EXPERIENCE

Present Sep 2026	Reader/Grader , University of Washington Information School IMT 521: Artificial Intelligence Foundations, <i>Instructor: Prof. Aylin Caliskan</i>	Seattle, WA
May 2025 Jan 2025	Reader/Grader , University of Michigan Department of Mathematics MATH 417: Matrix Algebra I, <i>Instructor: Prof. Ruiyuan Chen</i>	Ann Arbor, MI
Dec 2023 Sep 2023	Undergraduate Tutor , University of Michigan Math Learning Center Calculus I–IV, <i>Supervisor: Irina Arakelian</i>	Ann Arbor, MI

LEADERSHIP & SERVICE

Present Sep 2026	Global Sustainability Fellow , Impending Bloom Applied systems thinking and sustainability reporting frameworks to evaluate corporate disclosures, identify reporting gaps, and develop data-driven recommendations.	Remote
Aug 2026 Mar 2026	Program Lead , Develop for Good Oversaw four interdisciplinary student volunteer teams, coordinating project execution, client communication, and mentor engagement across nonprofit software development initiatives.	Remote
Feb 2026	Product Manager , Develop for Good — PainUSA Team	Remote

Oct 2025 | Led five undergraduate student volunteers to prototype a web application for PainUSA, featuring an interactive clinician lookup map and chronic pain resources.

SKILLS

Research Methods	Mixed-Methods Evaluation, Experimental Design, Statistical Modeling, NLP, Computer Vision, Information Retrieval
Programming	Python, R, SQL, Bash, JavaScript/TypeScript, Go, C/C++
ML & Data	PyTorch, scikit-learn, Pandas, NumPy, OpenCV, XGBoost, LightGBM, LangChain, Dask
Systems & Platforms	Git, Docker, GitHub Actions, Linux, GCP, AWS, Terraform, FastAPI, React, Node.js, MongoDB, IBM Db2